Finite State Machines

FOCS

Sept 21, 23, 2020

Last week

- Decision functions
- Formal languages = sets of strings
- Regular languages, regular expressions

Let's dig a little bit deeper into regular languages

How to check if a string is in a regular language?

We gave a definition of what it means for a string to match a regular expression

You can turn that into an algorithm directly, but it's not a very efficient algorithm - in part because of this rule:

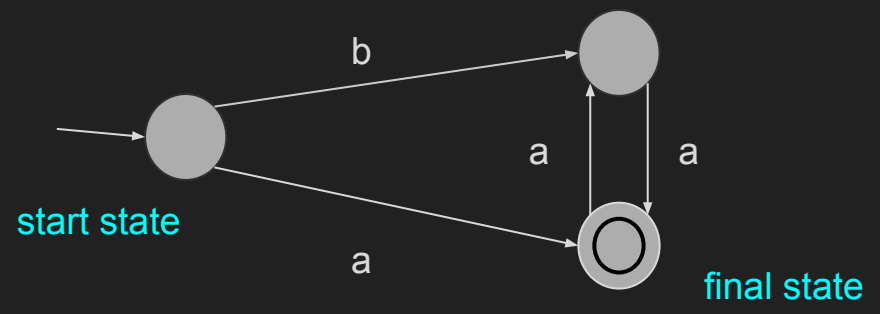
A string matches $r_1 + r_2$ if it matches r_1 or it matches r_2

Intuitively, this requires first checking that the string matches r_1, and if that check fails, then check if matches r_2.

(This is called a backtracking regular expression matcher, by the way)

Finite state machine

A more efficient way of checking matching for regular expressions is to use a finite state machine.



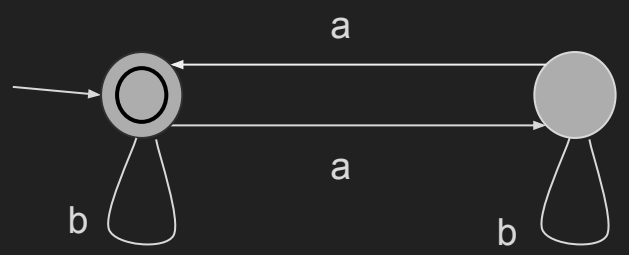
Finite set of states with labeled transitions between them (labels over alphabet Σ)

Given an input string, follow the symbols in the input strings starting from the start state — if you reach a final state, accept the input string; otherwise, reject

The above machine accepts ba, baaa, baaaaa, …, a, aaa, aaaaaa, ...

Examples

$\Sigma = \{ a, b \}$



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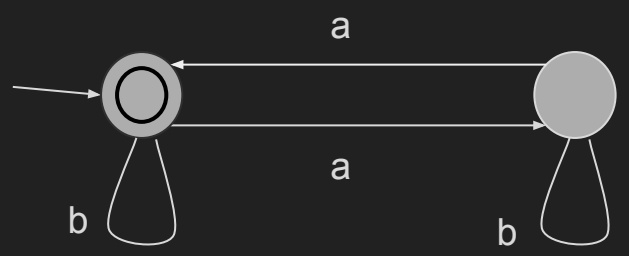
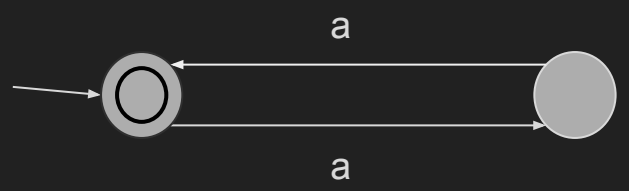


Accepts ε, aa, aaaa, aaaaaa, baa, baba, babbbab, …

Accepts all strings with an even number of a's and any number of b's

Examples

$\Sigma = \{ a, b \}$



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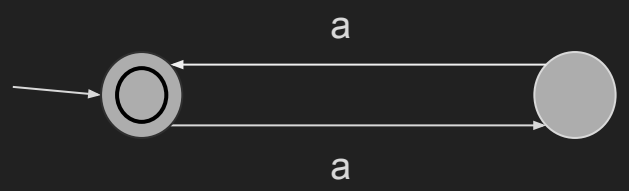
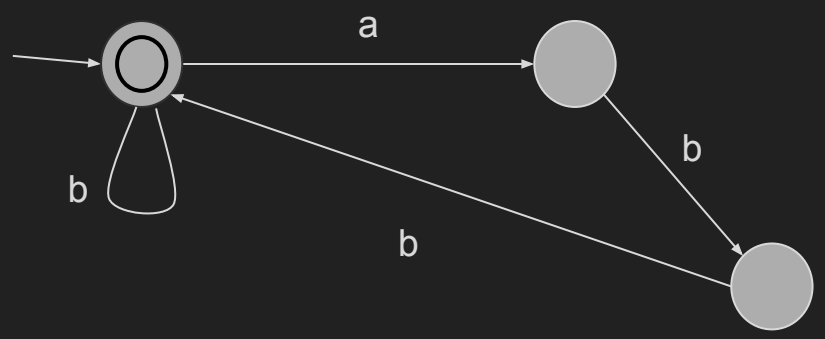


Accepts all strings with an even number of a's and no b's

Examples

$\Sigma = \{ a, b \}$



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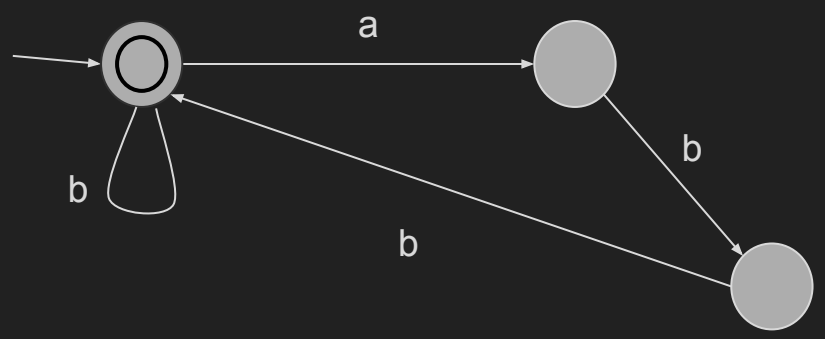
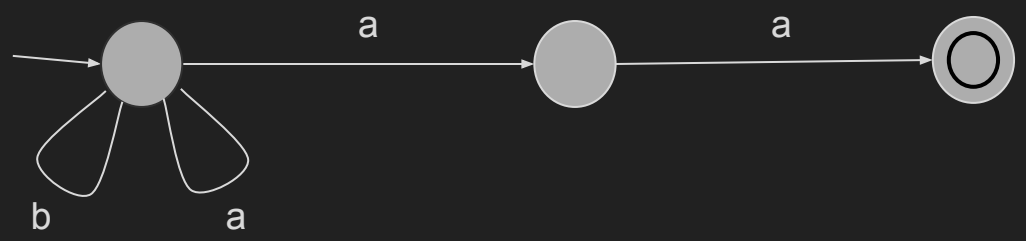


Accepts all strings in which every a is followed by at least 2 b's

Examples

A non-deterministic example (multiple choices)

Σ = { a, b }



Examples

A non-deterministic example (multiple choices)

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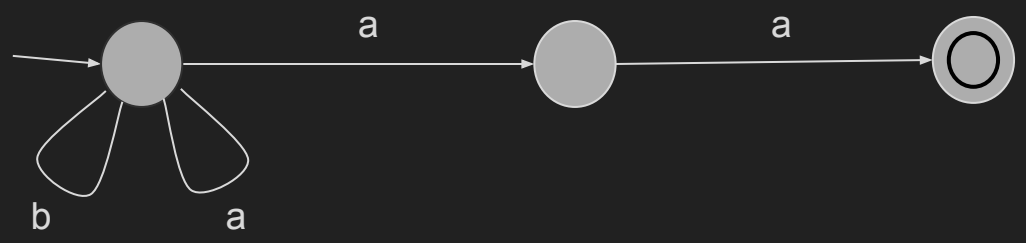


Accepts all strings ending with two a'

Formal definition

A finite state machine (or finite automaton) is a structure $M = (Q, \Sigma, \Delta, s, F)$ where

- Q is a finite set of states
- Σ is a finite alphabet
- Δ is a relation over $Q \times \Sigma \times Q$
  - $(q, a, p) \in \Delta$ when you can go from state q to state p by reading symbol a
- s is the start state ($\in Q$)
- F is a set of final states (each $\in Q$)

Accepting a string

A finite state machine $M = (Q, \Sigma, \Delta, s, F)$ accepts string $u = a_1 a_2 \ldots a_k$ if there exists a sequence of states $q_0, q_1, q_2, \ldots, q_k$ such that

- $q_0 = s$
- $(q_{i-1}, a_i, qi) \in \Delta$ for $1 \leq i \leq k$
- $q_k \in F$

Language of a finite state machine

The language L(M) accepted by finite state machine M is

$$L(M) = \{ u \in \Sigma^* \mid M \text{ accepts } u \}$$

Theorem: *A language can be accepted by some finite state machine if and only if it is regular*

We show one direction of the proof: if a language is regular, then there is a finite state machine that accepts it.

Regular expressions to finite state machines

Assume a language is regular. Then there is a regular expression that denotes it. If we can figure out how to transform any regular expression into a finite state machine that accepts the language denoted by the regular expression, we can construct a finite state machine to accept that regular language.

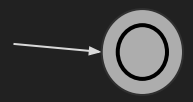
We define a recursive process that takes a regular expression and transforms it into a finite state machine that accepts the same language.

$$\text{build} : \text{RegularExpressions} \rightarrow \text{FiniteStateMachines}$$

Regular expression 1

Regular expression 1 denotes language {\epsilon}

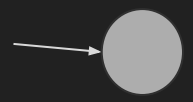
We can easily build a finite state machines accepting {\epsilon}



Regular expression 0

Regular expression 0 denotes language ((empty))

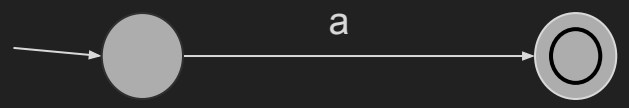
We can easily build a finite state machines accepting ((empty))



Regular expression a

Regular expression a denotes language { a }

We can easily build a finite state machines accepting { a }

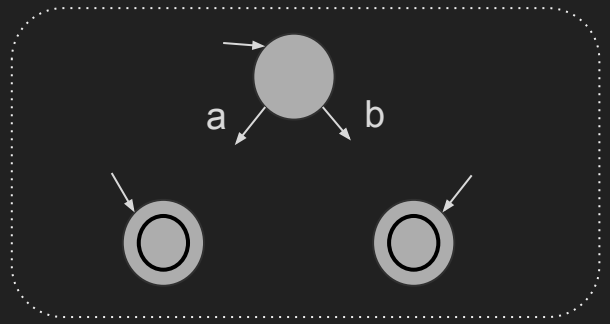


Regular expression $r_1 + r_2$

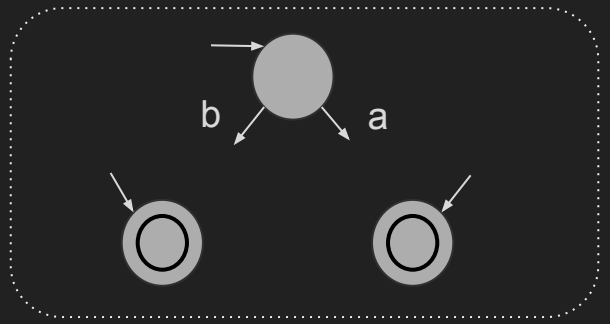
Regular expression $r_1 + r_2$ denotes language lang(r_1) \cup lang(r_2)

Recursively build M_1 that accepts lang(r_1) and M_2 that accepts lang(r_2)

Perform some surgery to build M that accepts lang(r_1) \cup lang(r_2)



M_1



M_2

Regular expression $r_1 + r_2$

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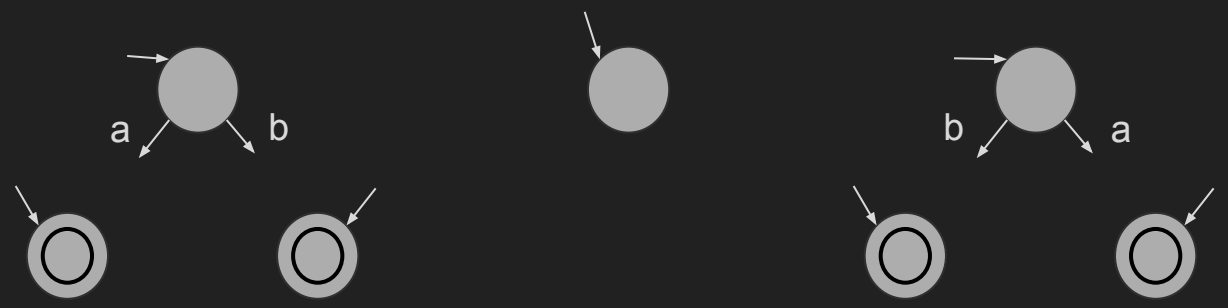

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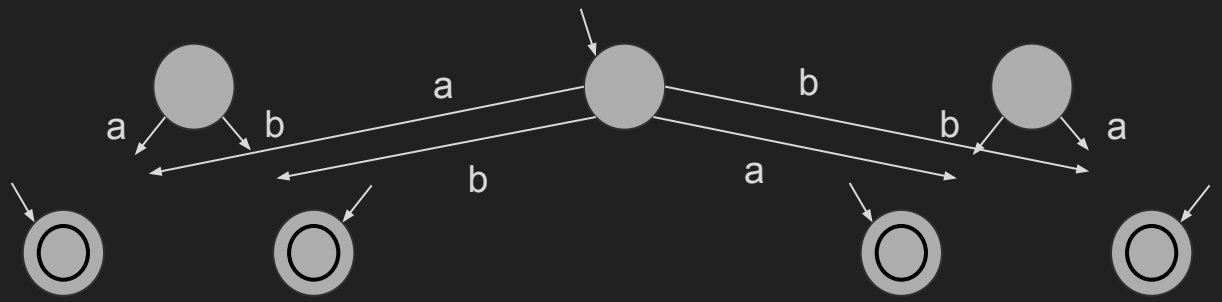

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Regular expression $r_1\ r_2$ denotes language lang(r_1) \cdot lang(r_2)

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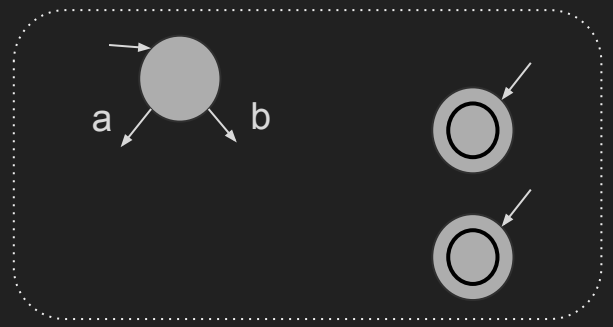
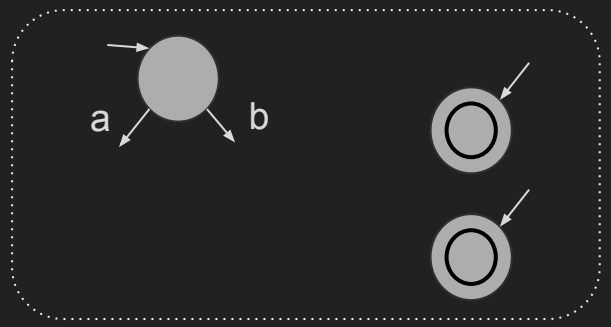
Perform some surgery to build M that accepts lang(r_1) \cdot lang(r_2)



M_1



M_2

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Regular expression $r_1\ r_2$ denotes language lang(r_1) \cdot lang(r_2)

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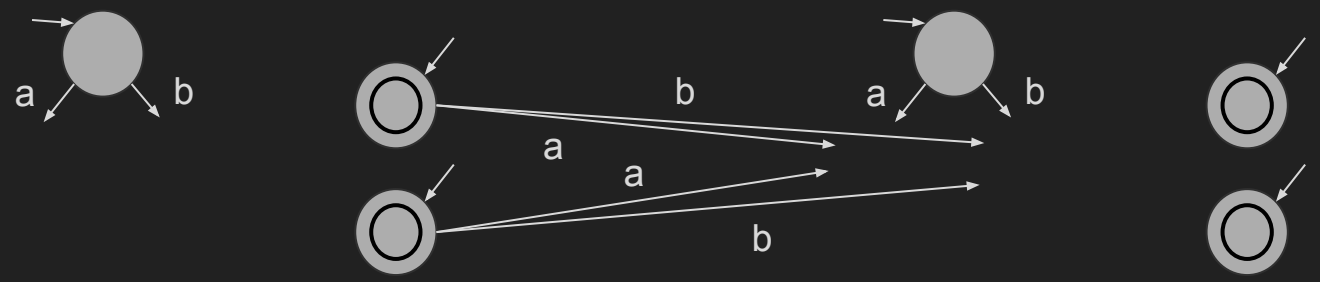

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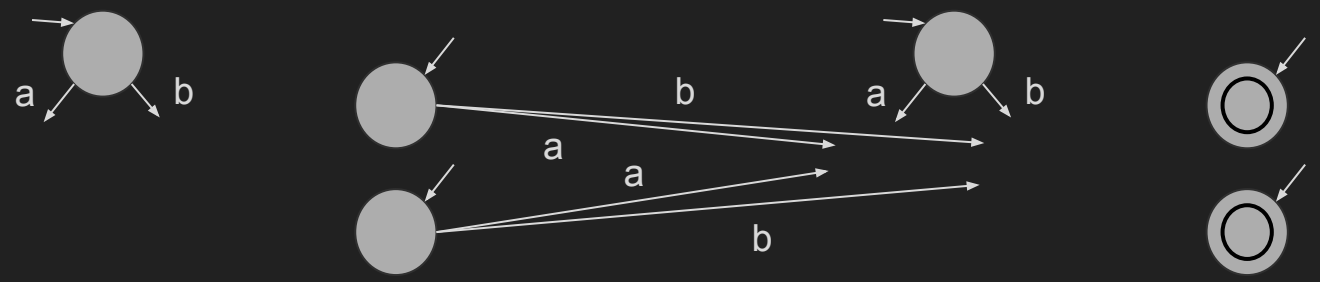


"Unfinal" these is s_2 is not final

Regular expression r_1*

Regular expression r* denotes language lang(r_1)*

Recursively build M_1 that accepts lang(r_1)

Perform some surgery to build M that accepts lang(r_1)*

Exercise - similar to r_1 r_2

Deterministic finite state machines

Deterministic finite state machines are finite state machines where there is at most one transition per symbol out of every state.

Sounds like a restricted form of finite state machine — but every regular language can be accepted by some deterministic finite state machine.

This is because we can always transform a finite state machine accepting language A into a deterministic finite state machine accepting language A.

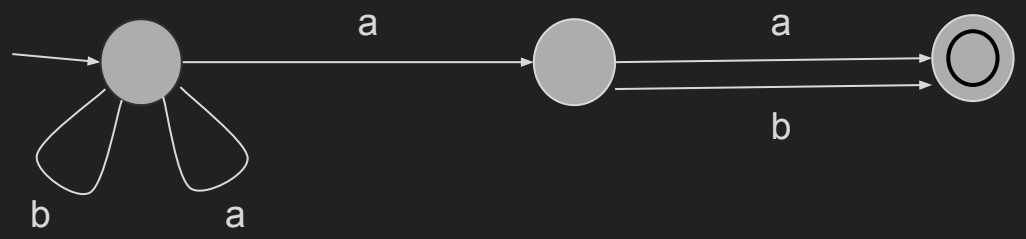
(Called the *subset construction*)

The subset construction

Idea: given a finite state machine M, create a new finite state machine M' where

- states of M' are *sets of states of M*
- there is an a-transition from P_1 to P_2 in M' if there is an a-transition in M from some state in P_1 to some state in P_2
- the start state of M' is {s} where s is the start state of M
- the final states of M' are all the states of M' containing a final state of M

Example



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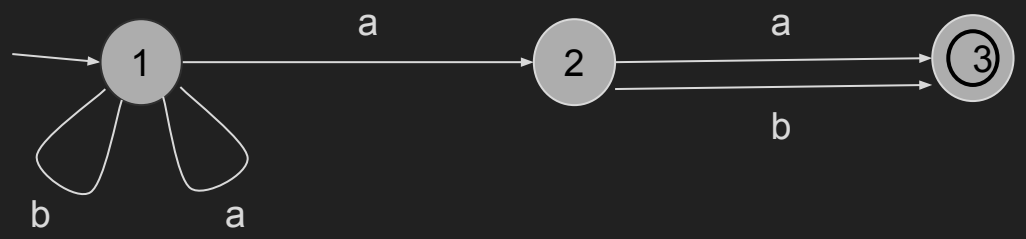

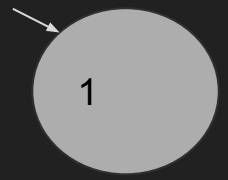

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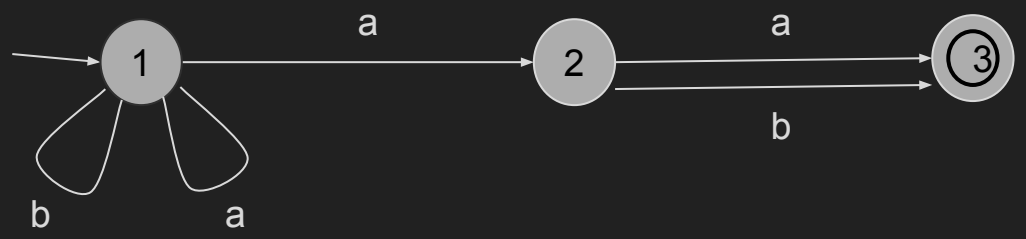

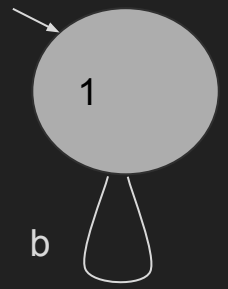

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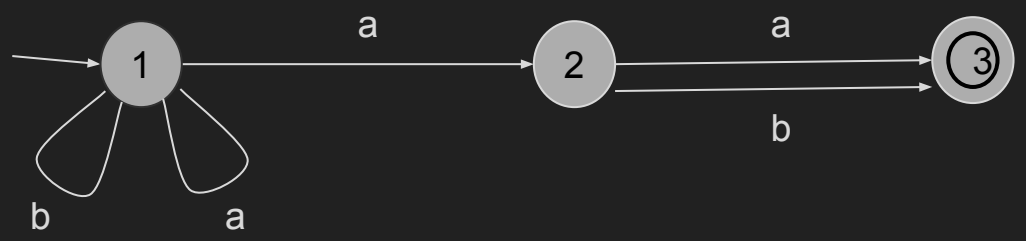

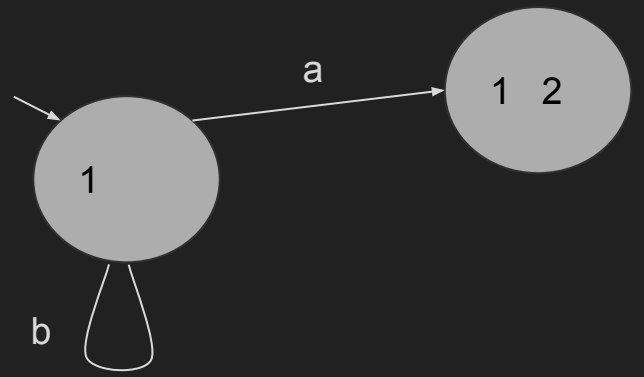

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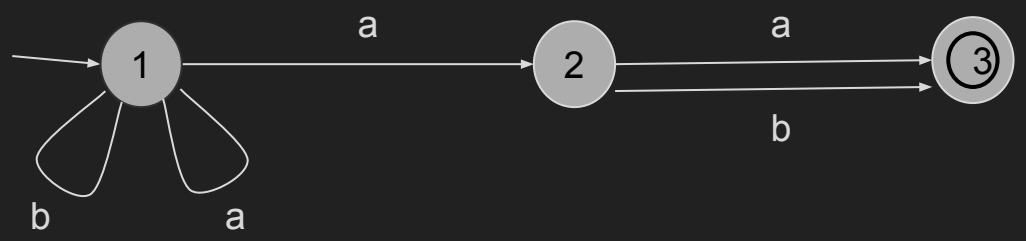

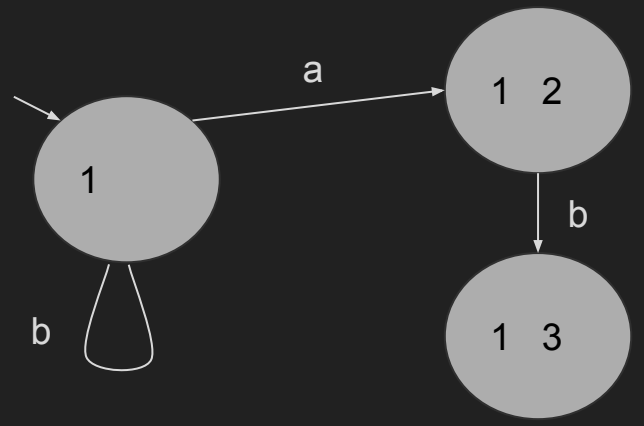

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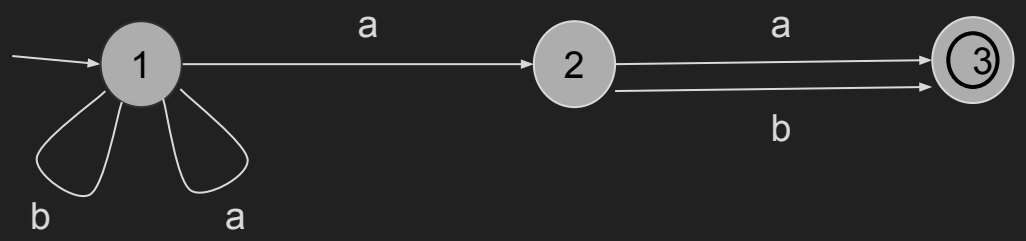

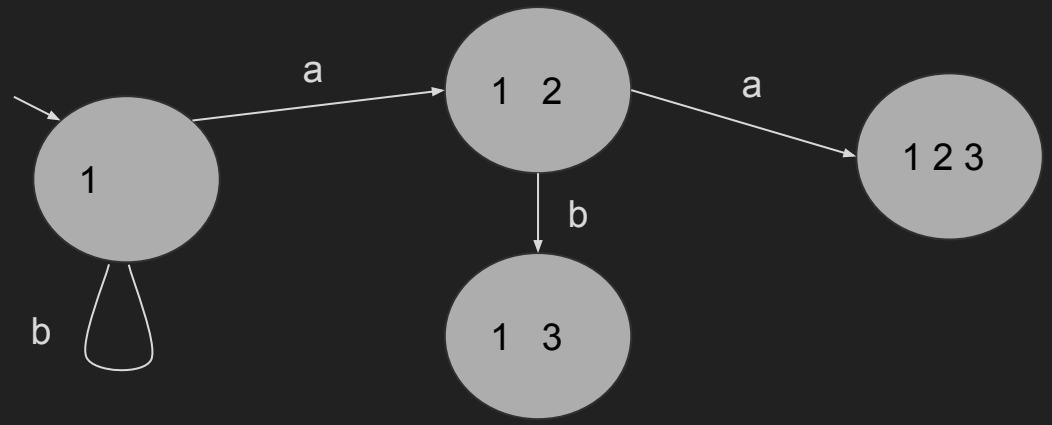

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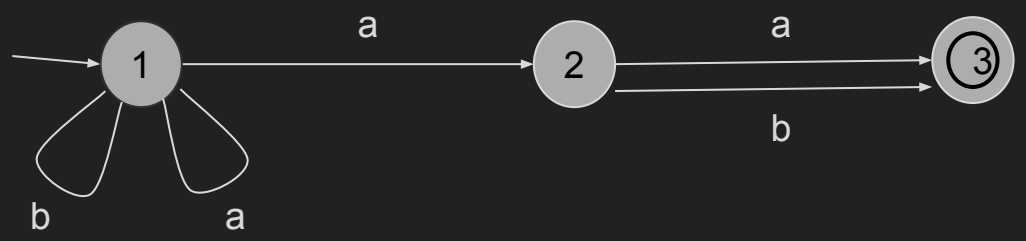

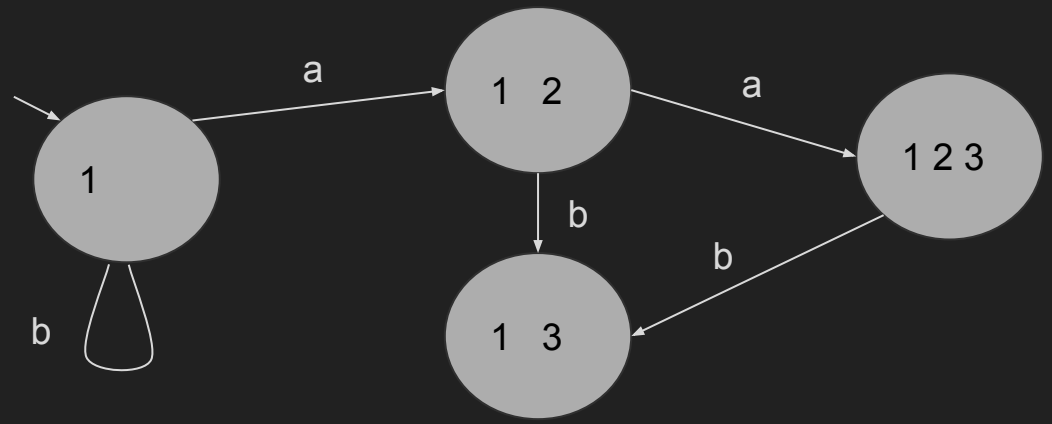

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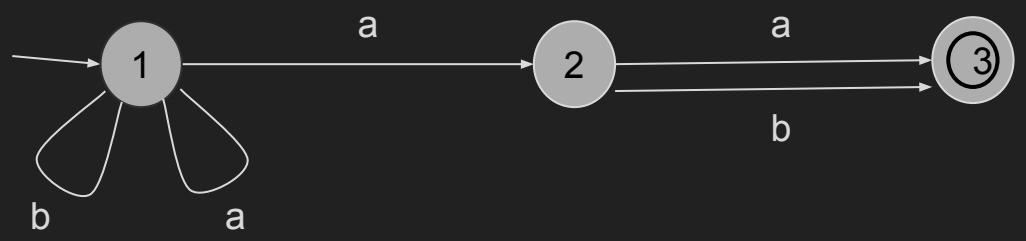

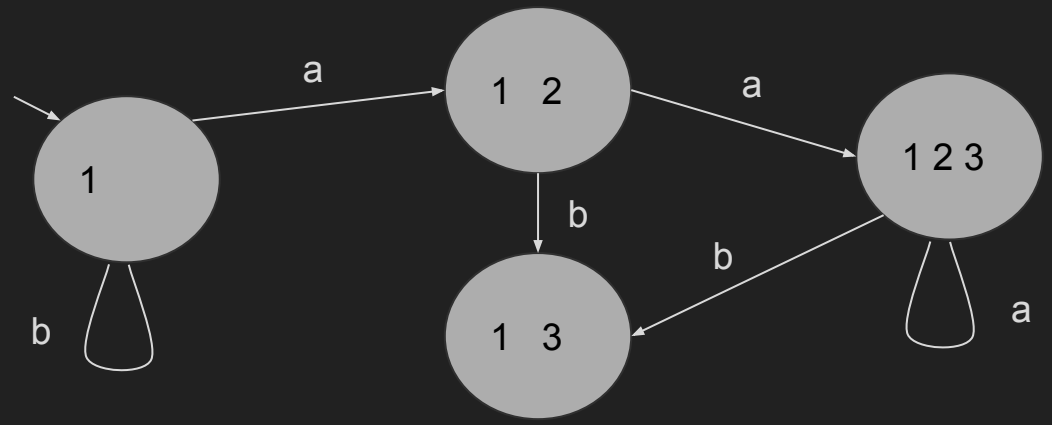

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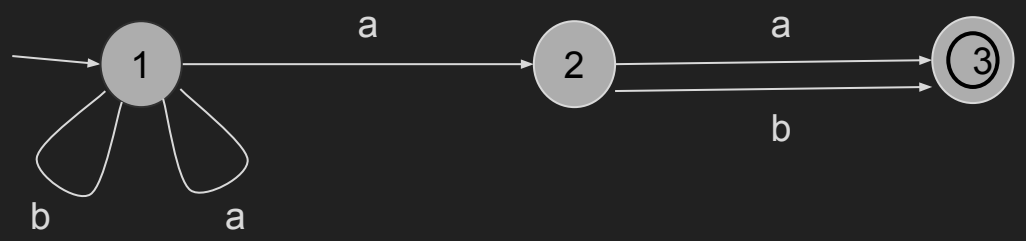

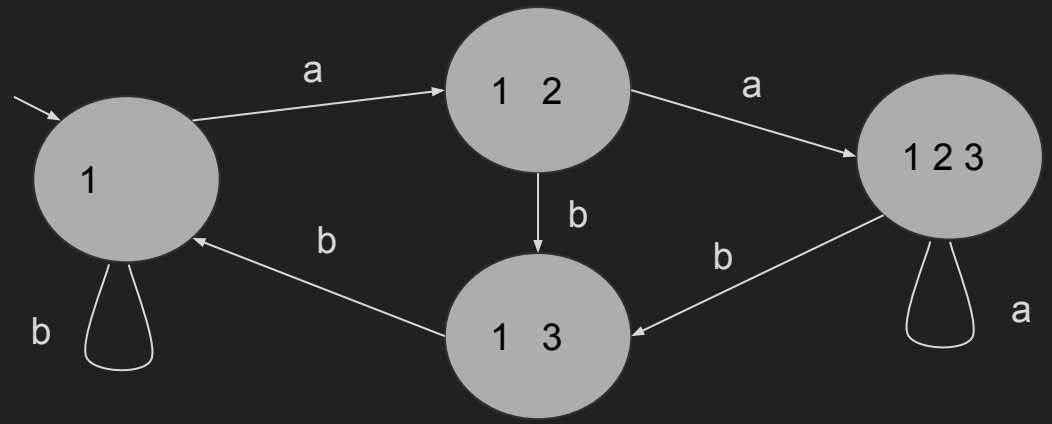

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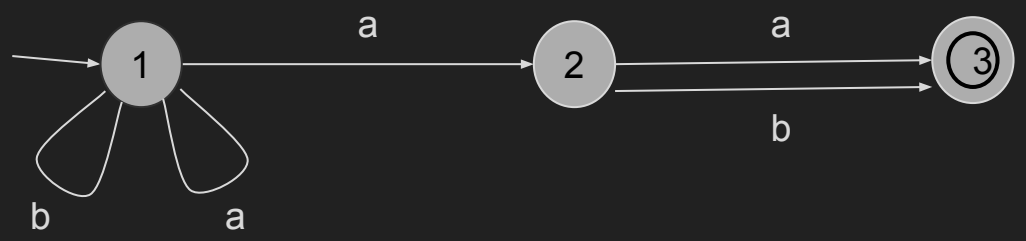

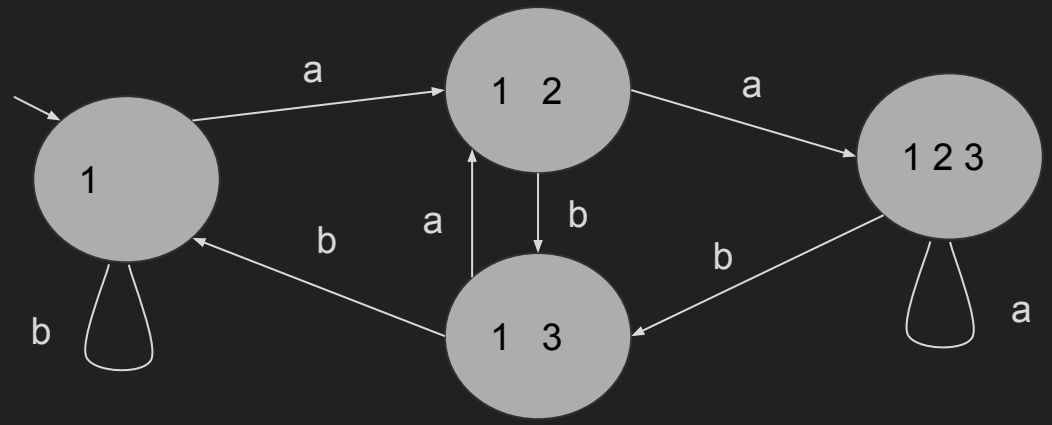

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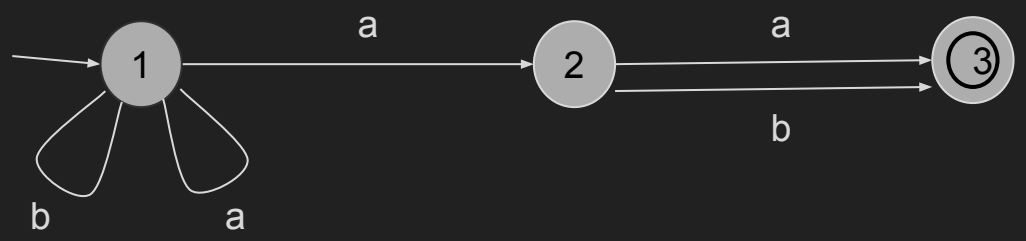

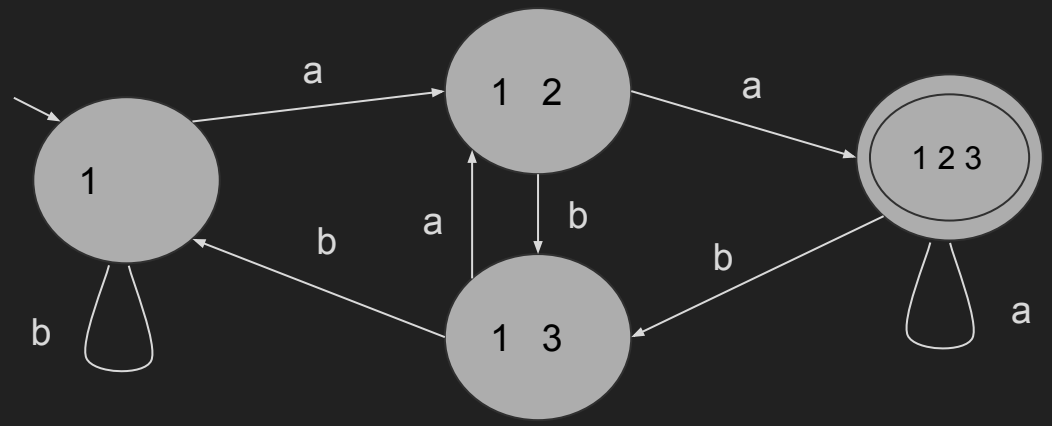

Finite state machines = our first computation model!

A finite state machine is an implementation for a function $f : \Sigma^* \rightarrow \{ \text{true, false} \}$

- f(u) = true when the finite state machine accepts u, and = false otherwise

A function $f : \Sigma^* \rightarrow \{ \text{true, false} \}$ is FA-computable if it there exists a finite automaton M such that f(s) = true exactly when M accepts s

Theorem: *$f : \Sigma^* \rightarrow \{ \text{true, false} \}$ is FA-computable if and only if its characteristic language A_f is regular*

There are many functions that are not FA-computable but intuitively should still be computable